

Datasheet POWER ATLANTIC 3 MWh



Energy Storage System (ESS) - 3.34 MWh

The ESS with a total capacity of 3.34 MWh is housed in a standard 20-ft container, with configurable power ratings of 1 MW or 2 MW.

The container switchboard includes an integrated UPS to ensure uninterrupted operation of control and protection systems.

The liquid-based thermal management system monitors overall system temperature and individual battery module temperature differentials.

The container is equipped with integrated HVAC, fire protection, and alarm systems, ensuring safe and reliable operation.

Num.	Parameter	Configuration	Description
1	Configuration	1P416S*8	8 battery modules connected
2	Rated capacity	3344 kWh	0,5P±3°C
3	Voltage range	1164,8-1497,6 V	Cell's voltage: 2,5 V - 3,65 V
4	Rated power/Charging and discharging current	1672kW/1256A	Standard charging current
5	Operating temperature	-30°C~50°C	Constant voltage charging mode
6	Communication system	Modbus TCP/IEC104/IEC61850	CAN(control unit)/ Ethernet(EMS)
7	Charging/Discharging Efficiency	93%	0,5P, 25±3°C, 90% DOD;80%EOL
8	Type of cooling	Liquid cooling	-
9	Auxiliary power	480VAC,50/60Hz	or 380/400AC, 50/60 Hz
10	Dimensions	6058x2438x2896mm	-
11	Weight	~31t	-
12	IP level	IP54	-
13	Lifetime	8000 cycles	0,5C, 25±3°C, 90% DOD;80%EOL
14	Self-discharging	<3,5% per month	All devices are switched off and the system is disconnected
15	Internal temperature	25±5°C	-



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High security

Vaporized fire extinguishing agent in PACK and container level, Additional water spray for container, Aerosol fire resistance for electric cabinet

Intelligentization

In-depth fusion of BMS/PCS/EMS software, support real-time failure diagnosis and prediction



Digitization

Digital twin system, supporting un-man operation and maintenance

High reliability

Intelligent comprehensive environmental control, support harsh environment operation

Low power consumption

Thermal resistance balance design and intelligent optimization control to support life time low consumption

Parameter	Unit	Value
Cell type	-	LFP 3.2V/314Ah
Battery configuration	-	1P416S*8
Rated Capacity	MWh	3,34
DC Voltage Range	V	1164,8-1497,6
Dimensions(W*G*H)	mm	6058x2438x2896
Weight(with/no battery)	t	~31/9
Anti-corrosiove level	-	C3
IP level	-	IP54
Operating temp. level	°C	-30°C~50°C
Humidity	%	0-95
Cooling type	°C	Liquid cooling
Altitude	m.n.p.m	2000
Work habitat	-	Outdoor
Extinguishing	-	Aerosol, combustible gas detection + ventilation, aqueous solution
Communication	-	Modbus TCP/IEC104/IEC61850



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Liquid cooled PACK



Parameter	Unit	Value
Cell type		LFP
Configuration		1P104S
Capacity	Ah	314
DC Rated Voltage	V	332,8
Rated Energy	kWh	104,49
Operating temp. range	°C	-30°C~50°C
Balancing mode	-	Passive/active balancing
Cooling type	-	Liquid cooling
Efficiency	%	>93
Dimensions(W*G*H)	mm	(790±5) *(2180±5) *(252±5)
Weight	kg	~690±10
IP level	-	IP67
Type of battery	-	LFP
Rated capacity	Ah	314
Cell's rated voltage	V	3,2
Weight	kg	5.6±0.3



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The cells used in the system are prismatic lithium iron phosphate (LiFePO₄) cells with a capacity of 314Ah and a nominal voltage of 3.2V. All cells are class A, and the operating voltage range is from 2.5V to 3.65V.

The self-discharge of the cells is ≤3% per month.

Num.	Parameter	Value	Remarks
1	Rated capacity	314 Ah	0,5P/0,5P; 25 ± 2°C; 2,5 V - 3,65 V
2	Rated energy	1004,8 Wh	-
3	Rated voltage	3,2 V	-
4	Charging voltage	3,65 V	-
5	Discharging voltage	2,5 V with T>0°C; 2 V with T<0°C	-
6	Max. charging power	0,5P	25 ± 2°C
7	Max. discharging power	0,5P	25 ± 2°C
8	Internal initial resistance	0,18 mΩ ± 0,05 mΩ	AC, 1kHz
9	Round Trip Efficiency	94%	25 ± 2°C, 0,5 P
10	Lifecycle	8000 cykli	25 ± 2°C, 0,5 P, 70% EOL
11	Weight	5.6±0.3 kg	-
12	Altitude		
13	Dimensions(W*G*H) [mm]	(207±1.0) × (174 ±1.0) ×(71.7±2.0)	
14	Distance between terminals	123,0 mm ± 0,3 mm	
15	Charging temperature	0-60 °C	
16	Discharging temperature	-30 - 60 °C	
17	Self-discharging	≤3,0% / month	storage in 25 ± 2°C



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The battery module is designed in a 1P104S configuration, using single lithium iron phosphate (LiFePO₄) cells with a voltage of 3.2 V and a unit capacity of 314 Ah.

They form a module with a voltage of 332.8 V and a total stored energy of 104.49 kWh. Each module has a separate Battery Management System (BMS), equipped with an individual extinguishing system that activates in the event of a malfunction (e.g., a significant increase in the temperature of the cells inside the module).

Num.	Parameter	Specification	Remarks
1	Configuration	1P104S	
2	Rated energy	104,49 kWh	0,5P @ 25 ± 3°C
3	Rated voltage	332,8 V	
4	Range of voltage	260 V - 379,6 V	Cell's voltage: 2,5 V - 3,65 V
5	IP level	IP67	
6	Charging power/current	50,92kW/157A	0,5P, 25 ± 2°C
7	Discharging power/current	50,92kW/157A	0,5P, 25 ± 2°C
8	Operating temperature	Charging: 0 - 55 °C	Discharging: -20 - 55 °C
9	Ambient temperature	20 - 30°C	-
10	NTC	30 pcs/module	-
11	Charging/Discharging Efficiency	≥93%	0,5P @ 25 ± 3°C
12	Self-discharging/month	≤3,0% / month	No auxiliary power
13	Cooling type	Liquid cooling	-
14	Communication	CAN2.0	-
15	Balancing method	Active/Passive	-
16	Auxiliary power system	24VDC	-
17	Auxiliary power consumption	0,5 W	Apart from cooling power
18	Connection type	Fast connector	-
19	Dimensions (W x D x H)	790 x 2180 x 250,5 mm	Tolerance: ± 3 mm
20	Weight	~690±10 kg	-



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Battery cabinet consists of 8 battery modules connected in series and 1 HV Box module, which contains, among others, contactors, fuses, a main switch, and a BMS module managing the operation of a single cabinet. The BMS module receives the voltage from the series-connected battery modules and communication signals from the BMS of each module. The master BMS system collects data from 10 HV Boxes, implementing a three-level management structure of the energy storage unit. The current outputs of each HV Box are connected to the HVDC switchboard, which also includes the terminals of a bidirectional inverter operating the energy storage system.

Num	Parameter	Specification	Remarks
1	Configuration	1P416S	
2	Rated energy	417,996 kWh	0,5P @ 25 ± 3°C
3	Rated voltage	1331,2 V	
4	Range of voltage	1164,8 - 1497,6 V	Cell's voltage: 2,5 V - 3,65 V
5	IP level	IP67	
6	Charging power/current	208,998kW/157A	0,5P, 25 ± 2°C
7	Discharging power/current	208,998kW/157A	0,5P, 25 ± 2°C
8	Operating temperature	Charging: 0 - 55 °C	Discharging: -20 - 55 °C
9	Ambient temperature	20 - 30°C	-
10	NTC	30 pcs/module	-
11	Charging/Discharging Efficiency	≥93%	0,5P @ 25 ± 3°C
12	Self-discharging/month	≤3,0% / month	Apart from auxiliary power
13	Cooling type	Liquid cooling	-
14	Communication method	CAN3.0	-
15	Balancing type	Active/Passive	-
16	Auxiliary system	24VDC	-
17	Power of auxiliary system	<40 W	Apart from cooling power
18	Connection method	Fast connector	-
19	Dimensions(W x D x H)	790 x 1140 x 2333 mm	Tolerance: ± 3 mm
20	Weight	3,2 t	-



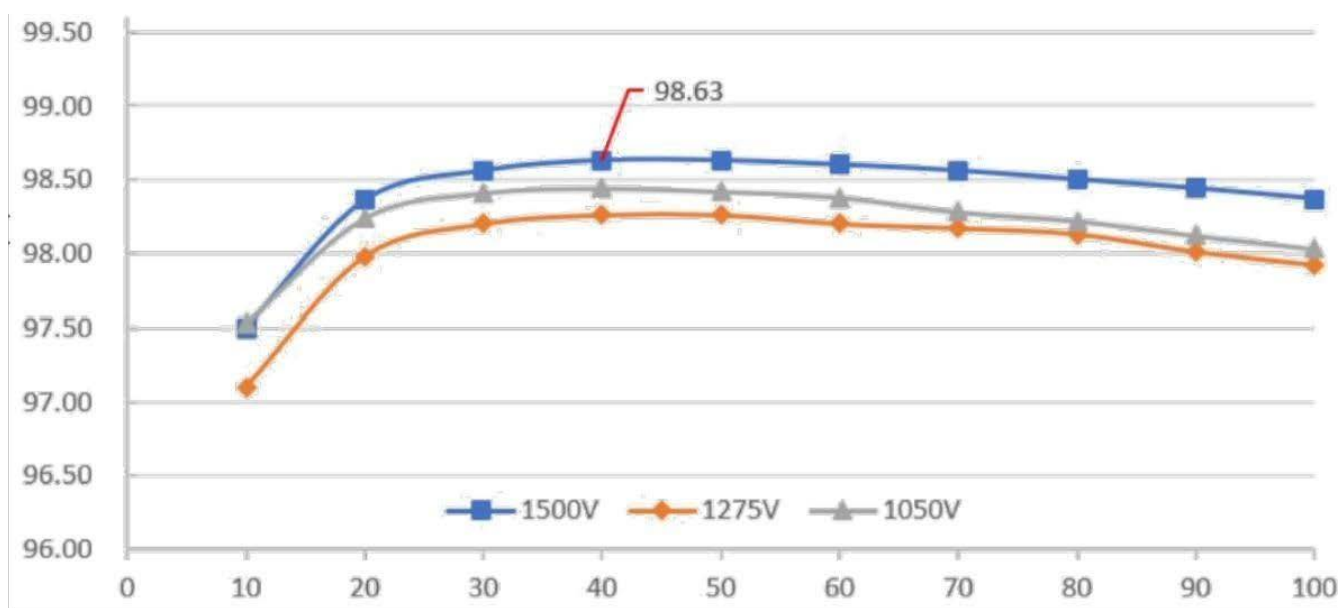
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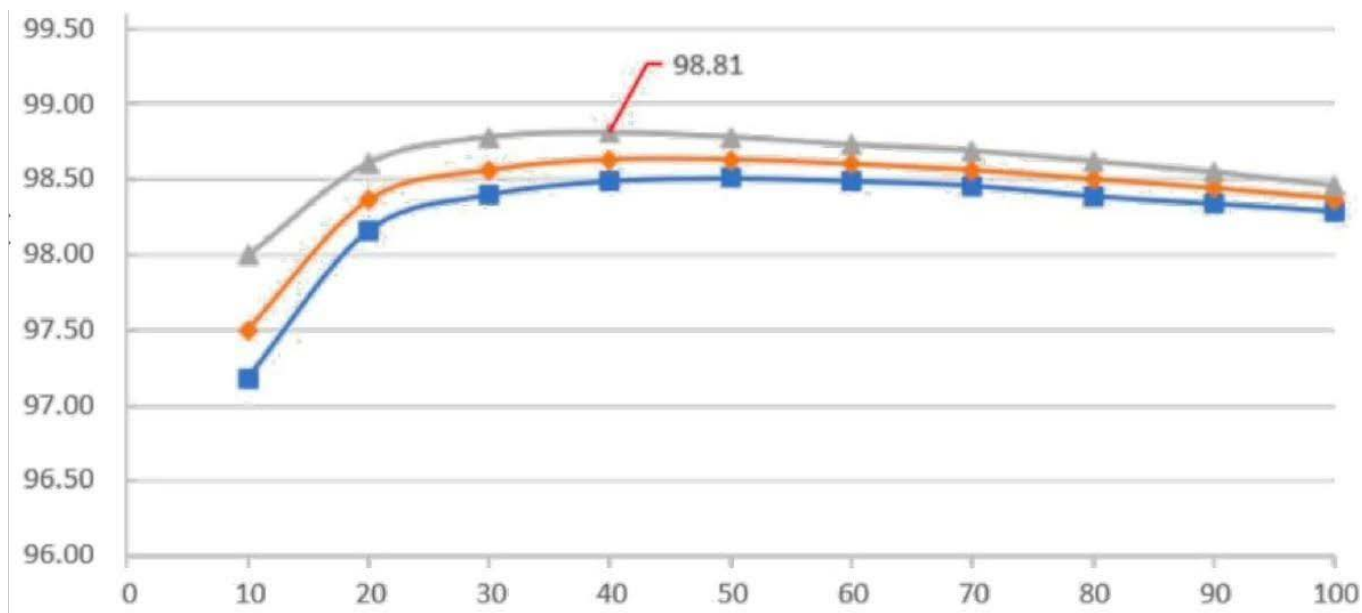
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Round-trip efficiency characteristic of the ATLANTIC 3.34 MWh energy storage unit as a function of active power



Rys.1 Efficiency characteristic during charging



Rys.2 Efficiency characteristic during discharging



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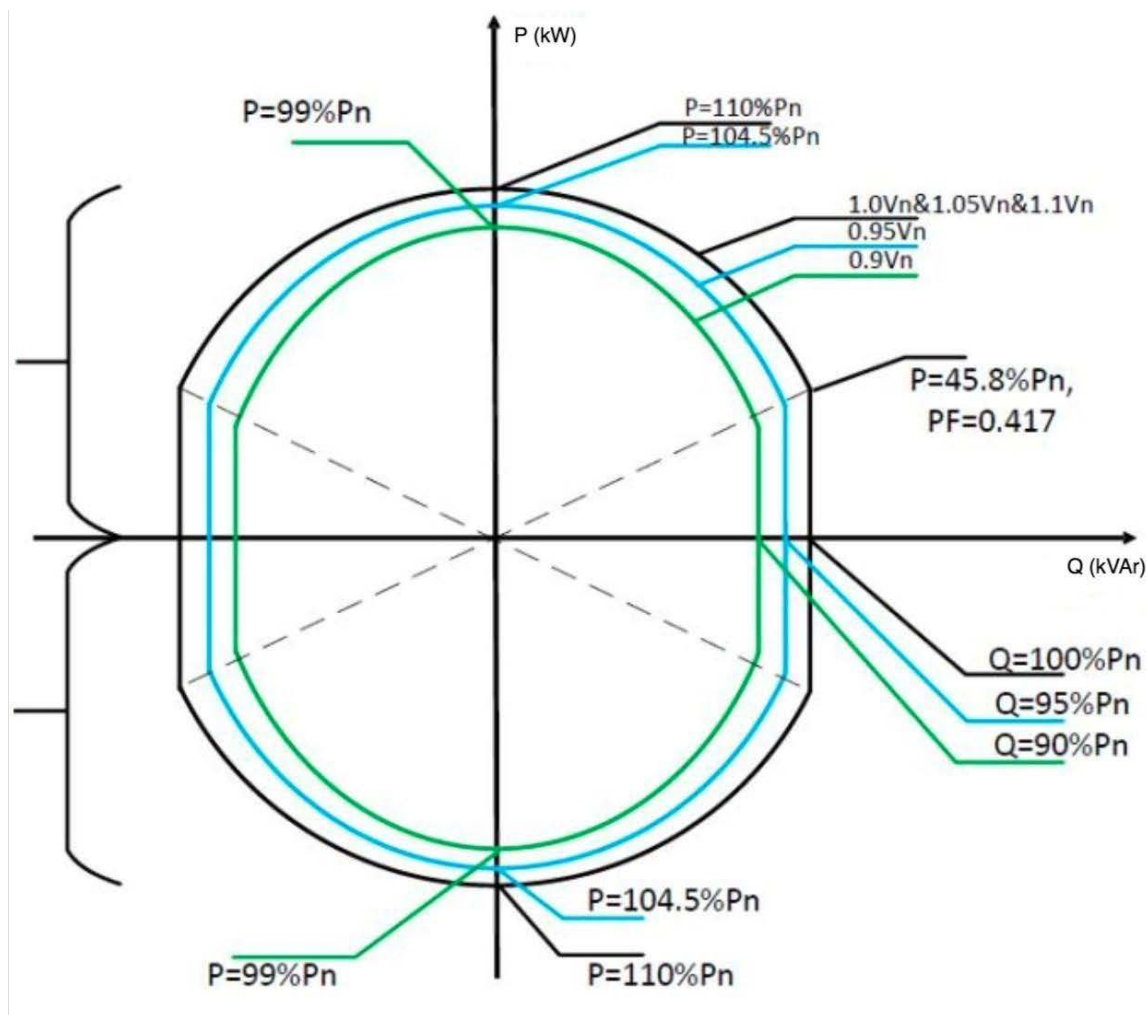
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Availability of reactive power generation and absorption over the full active power range in charging and discharging modes at the point of connection of the ATLANTIC 3.34 MWh energy storage unit



Operating area of the ATLANTIC 3.34 MWh energy storage system on the P-Q plane

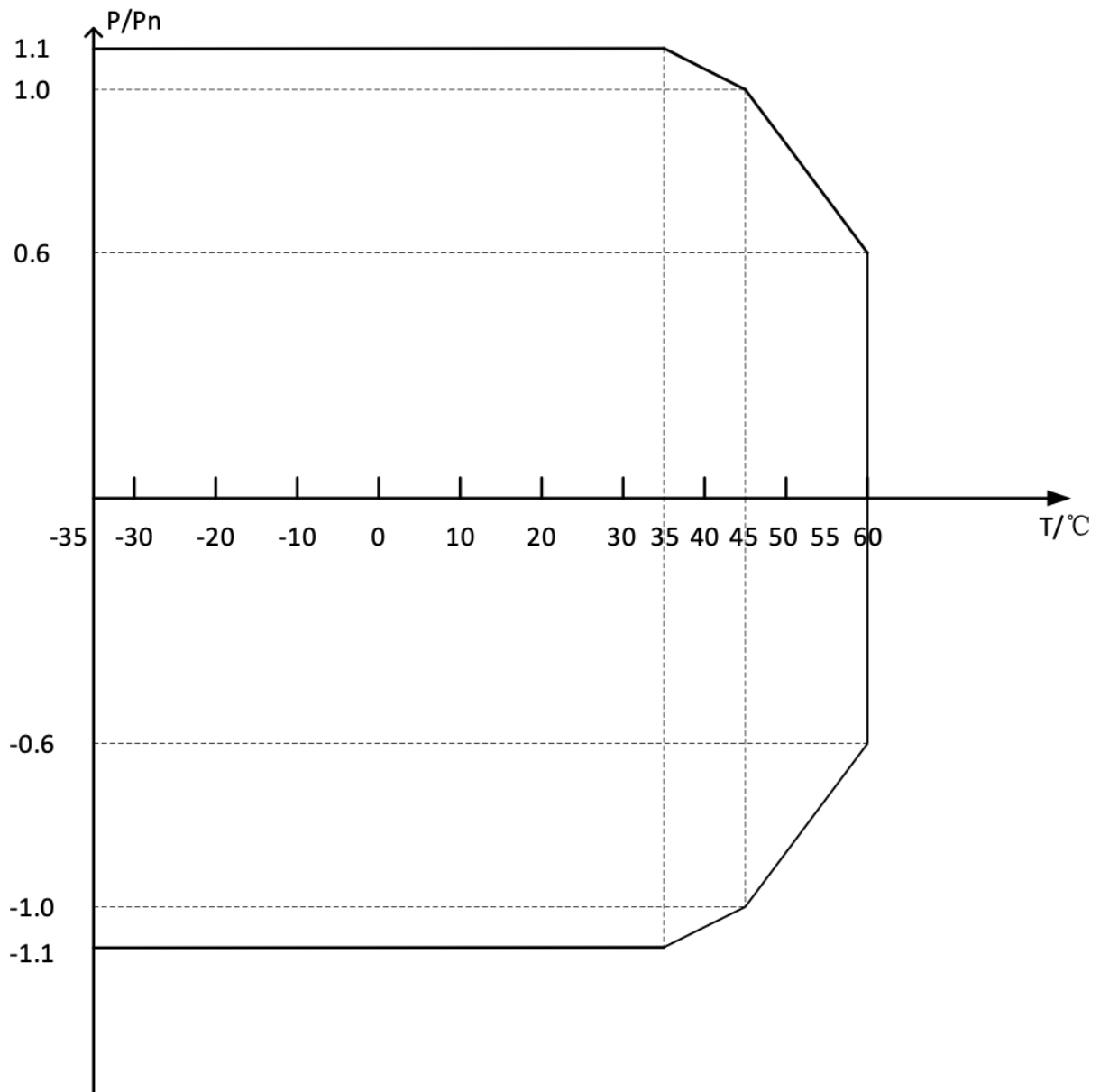


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Operating characteristics of the ATLANTIC 3.34 MWh energy storage unit taking ambient temperature into account



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